



Some people think that philosophy is of no relevance to everyday concerns.

I hope that our class has disabused you of this notion. However, in this last class, I would like to turn the tools of philosophical analysis on two questions of great current concern.

Was the College Football Playoff  
committee's decision to exclude  
Notre Dame competent, or  
incompetent?

Was the College Football Playoff committee's decision to exclude Notre Dame honest, or corrupt?

Logically that there are four exclusive and exhaustive possibilities.

The decision  
was  
competent  
and corrupt.

The decision  
was  
incompetent  
and honest.



The decision  
was  
incompetent  
and corrupt.

Here is an argument against the first possibility.

1. If the committee's decision was competent and honest, then the committee did not rely on absurd principles in making its decision.
2. The committee relied on absurd principles in making its decision.

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C. The committee's decision was not competent and honest. (1,2)

The argument is valid, and the first premise is obvious. It remains therefore to demonstrate the second premise.

The committee's task is to rank the country's football teams from best to worst, based on their performance. This is a general kind of task. One might also seek to similarly rank the best universities, or best philosophy departments, or best cars, or whatever.

One may rely on various principles in constructing rankings of this kind, and the appropriate principles will of course depend on the ranking. Nonetheless we can isolate one general kind of principle which one might use, which I will call a **magic touching rule**.

A magic touching rule is a rule which says that when two of the ranked items are 'touching' — i.e., adjacent in the rankings — then we look to a special extra condition — this is the magic part — to see whether the two touching items in the ranking should have their spots flipped.

## Magic touching rule

If two teams in the ranking are touching and played head to head and the lower ranked team defeated the higher ranked team then the teams' rankings are flipped.



The committee employs a rule of this kind.

It is easy to argue that magic touching rules are, in general, dumb. Either the extra condition (in this case the head to head result) is relevant to the ranking or it is not. If it is, it should be incorporated into the original ranking, in which case the flip is 'double counting' the relevance of the extra condition. If it is not relevant, it should also not be relevant in the special case of touching.

Further absurdities follow from the fact that the extra condition (winning a head to head matchup) is not a transitive relation.

Suppose that we have the following results: Team A beat Team B, Team B beat Team C, and Team C beat Team A.

Suppose now that we arrive at the following initial rankings:

Here is a second argument against the view that the decision  
was competent and honest.

1. If the committee's decision was competent and honest, then the committee did not inconsistently apply its own principles.
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Now we face our most difficult question. Was the committee incompetent and honest or incompetent and corrupt? In our current state of evidence, we do not have an argument which decisively settles this question.

However, we have already discussed a mechanism for arriving at probabilistic beliefs in such cases: Bayes' theorem.

Recall that in order to determine the probability of some hypotheses given the evidence, we need two pieces of information: the prior probabilities of the theories, and the probability of the evidence given the theories.

$$P(\text{CORRUPT}) = 0.5$$

$$P(\text{HONEST}) = 0.5$$

Because Notre Dame's exclusion required both events to obtain, we can set the probability of both happening in the absence of corruption to the product of their individual probabilities. So:



$$P(e \mid \text{CORRUPT}) = 0.8$$

$$P(e \mid \text{HONEST}) = 0.015$$

Real-World Bayes' theorem:

## Bayes' theorem

$$P(h \mid e) = \frac{P(h) * P(e \mid h)}{P(h) * P(e \mid h) + P(\neg h) * P(e \mid \neg h)}$$

It is now just a matter of arithmetic to compute the probabilities of our  
two theories.

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